

1. **Experiment:** Rosetta and MD for ABC Selection

2. **Time:** 2025.12.01-2025.12.17

3. **Member:** Ziqian He, Kaili Jin, Yulongzi Piao, Yue Chao, Haochen Xu, Qichang Liu, Xinru Yuan

4. **Method:**

(1) System Preparation

- ① The initial binding conformations of the designed protein-IL-6/sIL-6R α complexes were derived from our previous virtual screening and directly subjected to Rosetta energy minimization and molecular dynamics evaluations.
- ② Models satisfying the criteria of RMSD < 10 Å, Binder PTM > 0.7 or PAE < 5 were selected and subsequently subjected to Rosetta energy minimization, followed by further evaluation using Rosetta statistical potential (ddG) calculations.
- ③ The complex is placed in a periodic cubic water box, ensuring a minimum distance of 12 Å between any protein atom and the box boundary to provide sufficient solvation. The solvent environment is constructed using the TIP3P water model, with Na⁺ and Cl⁻ ions added to neutralize the system and simulate physiological salt concentration.
- ④ The protein is parameterized using the CHARMM36m force field, the water molecules use the TIP3P model, and ion parameters are derived from the CHARMM force field.

(2) Molecular Dynamics Simulation Settings

- ① All molecular dynamics simulations are performed using the GROMACS 2022.5 software package.
- ② We first ran energy minimization using the steepest descent algorithm to remove any unfavorable atomic clashes.
- ③ For equilibration, we applied a two-step procedure. In the NVT phase (1 ns), we restrained the protein backbone with a force constant of 1,000 kJ mol⁻¹ nm⁻² to let side chains and solvent relax. In the following NPT phase, we gradually reduced the backbone restraints and used Berendsen pressure coupling for initial pressure equilibration.
- ④ For the production run of the screening phase, we switched to the NPT ensemble with the velocity-rescaling thermostat (T = 310 K) and Parrinello-Rahman pressure coupling (p = 1 bar). Short-range van der Waals and electrostatic interactions were cut off at 12 Å, and long-range electrostatics were handled with the PME method. We ran each system for 50 ns. The first 20 ns of each trajectory was treated as extended equilibration and discarded; all downstream analyses were performed on the final 30 ns.

(3) Data Analysis

- ① We used GROMACS built-in tools to process the trajectories. For stability assessment, we calculated C α RMSD separately for the receptor (IL-6/sIL-6R α) and the designed binder, rather than aligning the whole complex. We also computed RMSF per residue to pinpoint flexible loops, and counted interfacial hydrogen bonds (donor-acceptor distance $\leq$ 3.5 Å) and buried solvent-accessible surface area (BSA) to monitor interface persistence over time.
- ② We used VMD to visualize trajectories and extracted representative snapshots from the initial (t = 20 ns) and final (t = 50 ns) frames to compare binding-mode changes.
- ③ To capture global conformational shifts, we constructed relative free-energy landscapes based on the first two principal components (PC1 and PC2) from the C α covariance matrix, with $\Delta G = -k_B T \ln[P(PC1, PC2)/P_{\max}]$. We also extracted van der Waals and electrostatic interaction energies along the trajectory and plotted them as radar charts for cross-comparison of the four complexes, alongside B-factor profiles to evaluate residue-level flexibility.
- ④ Finally, we overlaid the initial and final binding interfaces to identify any loss or gain of key residue contacts during the simulation.

5. Result:

Table 1. Candidate sequences.

ID	Binder PTM	minimum inter-chain pAE	target- aligned binder RMSD	statistical potential	MM/GBSA	MM/PBSA	Binder Seq
part1_1p9m_11_model_9_1	0.83	9.47	2.11	-165.69	16.01	-30.18	MELIKLLKEIYKVVKDKEQKALILLLIGEEYYKLGLKEEA EKYFKEALKLAKEAEKEGELMKMIKESLEKVEKEK
part1_1p9m_0_model_5_1	0.81	12.98	3.12	-97.42	8.92	-28.55	MIENLKKLIEAAKKAKDPNLGYQYLVLIDFLLKENKKEEA KEVLEEAKKDPNVNEEDKKKMEELLK
part1_1p9m_12_model_3_1	0.64	3.53	7.38	-188.66	3.46	-32.58	SHPVEEKLEEVLLKKLEELEKEEEKKLKEELEKAKEKQKENP EEAKKKAAEELEKEIKKLEEEGEKLIIEELKK
part1_1p9m_3_model_4_1	0.7	21.46	4.05	-122.01	1.08	-29.86	EEELQKEVAEILPKYMKEKKDYNEALKAAAEDLNKEWVVK TPYKLVYDEETNEIVVVPKS
part1_1p9m_36_model_2_1	0.81	14.02	5.93	-154.41	0.66	-33.2	KRKEEIKKLVKELLEAVKNGDEEKAIEKLGGKKLKVPEQ YDKIEELVVKYEIAYKEGDDEAMKKYLKELEKAVEE
part1_1p9m_36_model_8_1	0.78	12.34	6.92	-135.24	-2.51	-44.28	DVETAVKVQTEILLDQLEEILKKKIEEGNPEEAKPKCEEA KKKLEEAKKKALKIENPEEAVKYLDELAKEEELEKLE
part1_1p9m_32_model_1_1	0.78	15.56	9.98	-133.44	-3.15	-33.6	ELKELEKEISEEIEKLWDILLKIDVEKMAELQEKMMDLSG IEAVEYLIIEYVSSLIDTEEVKKIHKLLKELLKKIKEL
part1_1p9m_6_model_1_1	0.8	15.83	6.12	-151.24	-3.51	-55.56	DEEEIKKLLLEEAKKALEEKKYEEAEKLAREVIEKSTDKEL IEEAAEILLIARLALAKSAEEAIEAVEEELEKLVKEKL
part1_1p9m_9_model_2_1	0.77	2.93	2.69	-230.48	-5.37	-28.91	SEEKEKLIEEKLKEAEELLKENVPAEVVIKMAEYELSKNP LLSEEDVKKAIERLKEKIEELE
part1_1p9m_28_model_2_1	0.75	10.74	7.76	-161.27	-5.97	-52.32	YEELAKKLKEKGFEEAAEVLEEYLKKGKLEEAKKEAEKAA AEELEEERKKALSIDEEEREKALEEVLKKEEELEKLL
part1_1p9m_24_model_6_1	0.71	6.87	1.64	-220.47	-6.3	-54.42	MEEAEKVAKKIVESALGPEIKKLEEKVKEYTEKGDFENAE KAKKELEEKKKKYEEELKKLLEEVKELL
part1_1p9m_6_model_4_1	0.71	2.71	3.91	-149.72	-6.88	-26.13	KELLKKKAEEAKKKAEEELKKKVEELKKKGTDSELEEAIKL RLEAAEKEFAKKLEELAKTVSEEEAEEAIEKLKKEL

SynthImmunol_NMU			Notebook			No.5	
part1_1p9m_24_model_5_1	0.7	11.06	6.4	-152.07	-8.88	-36.51	TRLLPVDEAAIAALVDEELAPIRASLATADPATAAAAYLAE LAAEGRAPLEAAAAAARAAGLTEFEVEL
part1_1p9m_30_model_0_1	0.72	13.75	7.85	-202.51	-9.32	-39.9	SMVEKQYEEHIELCKELNKPELAЕКVEKLKKEALEAEKSR DPEALIAKLEEMEKLYEEVKKLKEEE
part1_1p9m_28_model_7_1	0.63	4.43	2.12	-172.35	-12.57	-53	KIEEYIEKLKKLEEELKKAEEEGNEEKIKKLKKEKEELLK KKDEIAGEAAVELIKKMAKEGKDPEEIVEAIEEMEKKL
part1_1p9m_38_model_0_1	0.72	15.94	3.27	-143.36	-14.32	-46.61	MEEEFKKFEEELSEELFEKYKELAVKGKEEEALELEKKEF LKKVEEYLKEKGFSEEEVEKKIKELEENL
part1_1p9m_25_model_4_1	0.7	15.94	3.27	-148.92	-17.03	-34.55	MLEKAREFAIKILEKDGIPYANEEMIEALVEEVLKILEKN PNATFEEVYЕК
part1_1p9m_27_model_5_1	0.84	5.69	9.83	-169.4	-17.06	-39.71	ENEKKAKEAYЕКGKKYLEEKKYDEAIKYFEEAIKLNPNNSK IASESYYLIADIYMKEKKLPEALELLLKANAIPNQ
part1_1p9m_23_model_1_1	0.72	14.29	5.07	-164.08	-17.56	-53.94	EVNNHPEVKKLKEELEKAKAEGDEEKAKKLEEELEKKSFE LELLSLGLSKEEAEKLAELYIESLKKEKE
part1_1p9m_27_model_2_1	0.75	6.95	5.26	-220.85	-17.65	-51.5	EKGEKLLKEAEEHAKKAЕELKKEAKKTKDPKKKEELLKKA KEHLEKAEEALLKAMVYYPEEYEELLKKLEELEKI
part1_1p9m_17_model_9_1	0.81	8.32	6.57	-193.13	-18.23	-33.6	EVEELVKEAKKLMEEKKYDEAIEKIKKAVEKSDDPEIKDL YYEALLRAGKVVEEAQEFAEKNI
part1_1p9m_38_model_5_1	0.73	9.18	1.58	-241.22	-18.87	-42	KELSLFGKALLKCLETEKLPSEEIFDRAVVEAAFALSKER EGVIEALCKMTGASREEAEKLIKLGKYF
part1_1p9m_29_model_5_1	0.77	5.52	6.98	-165.91	-19.82	-64.44	SALEELLKKEAEELAKKFQEKYPVPNLEKVKEKVYKKLKE LEKKNEEISEEELIEELKKEL
part1_1p9m_6_model_5_1	0.7	10.25	6.9	-240.45	-21.11	-76.57	EVEELLKKIEELEЕКFLERVRAKDPEEREKIKEEAREEL KPLLEKLAELGGSPGEFTEKQLELAERLERVWKES
part1_1p9m_14_model_4_1	0.73	1.35	3.02	-282.39	-21.22	-62	SKVEELVKRIEEKEKELEKELEELKKKYPELTEEELEKIK EERSEEIIKELIDEATKES
part1_1p9m_31_model_1_1	0.75	18.22	9.19	-131.93	-21.87	-64.73	AGEQLEKFLEAIEKINEAIKKKDVEEVKKYMKELEKLREE VLEKLGPNSELVELAEKTLKEAKEYIKKEEEK
part1_1p9m_28_model_3_1	0.78	10.48	5.16	-245.35	-22.51	-53.79	ALEKAKELIKEALKLEEEGKYEEAIKLYKEAAELDPSLKG ECLFLISLCYLLGKRKEAAESLEKALPDLVKESPYDI

SynthImmunol_NMU			Notebook			No.5	
part1_1p9m_21_model_2_1	0.71	15.74	6.87	-197.21	-23.6	-41.08	MEELLKKAKEAIKEGNKEEVEKIVEEVIEKGDLESLEIY KECEKYGIE
part1_1p9m_31_model_4_1	0.76	13.36	2.27	-240.2	-27.15	-54.74	AAAEVEAAVARYEAAAERYLAACKRAGAITPEQERELARAY NELVALGDPAAAAARLAELDARFAAEEAAAAAA
part1_1p9m_29_model_7_1	0.78	16.37	3.84	-200.94	-31.61	-50.21	ELEKIFKEYKEIHKYKEGKLTKEEYEKELDKLLEEIQKL LKDDIEKQIEAFEELEKIEKE
part1_1p9m_39_model_6_1	0.79	1.88	5.06	-261.84	-32.26	-73.88	MTVEELKKEEIEKLEKEIEAAKAAGKPAAEAHLQEKLEVLK SFLKELEEDKENKEALSISLRGTLALIDKEAAGEL
part1_1p9m_13_model_2_1	0.71	12.53	6.36	-152.09	-34.32	-57.06	MDVEKKIEEIEKIEKEAEKAKKEAEKYKEEAKKLKEEKAP EEKVKELEKKAKEAEKEAKKLEEEIFKYLKEAAV
part1_1p9m_24_model_7_1	0.81	1.47	5.53	-237.87	-34.34	-75.9	MELEKLIKEIFEKVEKYLKLKKENASEEELKKLEEEIEKL VEEAYKAAKKEGKLAKFQEALLEIDKLL
part1_1p9m_6_model_8_1	0.77	1.32	1.55	-286.37	-34.9	-62.42	TKEEELKKEALKAAASFIVENKKIAPDLVDEKLEKALKDY IKEISEKFGKPKEEVEEIEKEIEKLEKIGGKVTEEA
part1_1p9m_20_model_8_1	0.82	4.7	4.08	-262.87	-37.38	-77.71	MEEELKKKAEEELAKKAEEAREKGDKKKAIEYLEELLKVLN EIKTDPEKEEKIKIAEVELAWEKGDWKAYLEALEKL
part1_1p9m_20_model_5_1	0.75	4.19	3.67	-232.63	-38	-56.23	MIEEEAEKLAKEELEKVYEEKFKELVLKGPEFAIEEVEKL AKEEAKKKIIEYLIEEKKIPKEEAEDIAEKVLEKLK
part1_1p9m_37_model_7_1	0.72	1.2	7.38	-252.22	-40.7	-72.52	KEELEKEMEKIMKEIEEAEEKGDKEKARELMKKLIELEEK YLK
part1_1p9m_0_model_6_1	0.65	1.86	1.45	-279.13	-43.78	-76.75	IDPEKLKEEIKKLKEEKLEEFEPKYKELIKELAKKNLPEE EVLKEVEKFYKEFEKEFEKEIEKLKK
part1_1p9m_4_model_3_1	0.83	3.1	2.72	-255.47	-50.57	-70.4	SKEIEKEAEKAFEEKKYEEAIKLYEEALKKDPSKKLYYKE GIAFAYYELGKKKEARKLFIEIAEEL
part1_1p9m_0_model_7_1	0.73	14.78	4.69	-195.82	-50.67	-78.66	DLEVEMLKRLAEEAEKEGNWEELEGIGLELKKLGYEEEA KVLERAIKGYKEKGLEEKAKKVEEIL
part1_1p9m_20_model_1_1	0.79	3.08	5.91	-207.5	-53.73	-90.38	ILEKIKEYIKEKLRKGEVISLEELWKIAKEKFGITKEEAE EELKEAHEEAVFEVIREKAKEGASVEELIEFLEKAE
part1_1p9m_4_model_0_1	0.77	17.47	1.43	-207.73	-56.17	-92.92	EEEEKRKLKREYQELIIEYEKAE EEGKIEEAVKLLEEILK LAEKLGLPEEEIKSIKEELEKLKKKI

SynthImmunol_NMU			Notebook			No.5	
part2_trp249_1p9m_33_model_0_1	0.77	17.68	9.14	-78.43	-0.23	-11.62	LAELYGAAQRALAEERYEEAIALLEEILARREEVPPSLL AAAERLLAEAREKL
part2_trp249_1p9m_19_model_3_1	0.75	18.74	3	-155.85	-3.89	-25.71	ETVTTEVVVEEPDRLTITRTVSGVAEPPPERIEELQRALEA EVRARGFEPVLVRAVVENGVLTVTAVGVRPL
part2_trp249_1p9m_25_model_4_1	0.79	19.03	5.06	-105.19	-8.68	-27.21	MKKA EVEIYDEDGTLVHSEVVPVAEDGTVTYTFTVPES GKYTVKVYDEFGRLVYEETIEAKAGETYTVTVKVEP
part2_trp249_1p9m_19_model_7_1	0.84	20.78	6.68	-95.55	-9.54	-24.2	EREALVERAVARYEEVRALLEAGDLEGFEEAIRRIAEEL GEAVATTIGALARNRAPAEEARRVIARVVAAL
part2_trp249_1p9m_34_model_4_1	0.71	12.18	5.76	-109.53	-9.99	-32.44	PELVGEAVLDGRRLRLYLRETKDEQGKPVWELEEFADG ELVRRRTFASREEALAHVARLRG
part2_trp249_1p9m_28_model_7_1	0.72	16.33	7.38	-96.18	-12.79	-35.54	LYEAVVTWEKNGQKGRAVRVYPSREEAEAAAEIIRRL EAEGATNIRVEIR
part2_trp249_1p9m_8_model_4_1	0.77	15.29	2.54	-164.05	-15.36	-24.36	VVRS AVSALAKTPEEALAKAKAKAEALGFKVLEVASLV KVGDKWQATVRGEKEV
part2_trp249_1p9m_14_model_7_1	0.71	21.44	3.48	-86.01	-15.87	-33.74	VLEVAAYSVSADGKTVVVNINQTAAALGISVEEAVARL TAAVPSTVIETAPDGT VYGRVPAAAFAA
part2_trp249_1p9m_36_model_5_1	0.74	7.73	9.36	-177.24	-16	-30.91	KTVERVVGEVVPAGTPAEERLAAVEALAAERAPAKPA RLRMELVEVL PDGSVRYRVIAIYEL
part2_trp249_1p9m_32_model_0_1	0.85	21.6	7.88	-131.7	-16.13	-25.69	QREAVVEELERYGISRELAERLVEEAAALGAASAVVSPE EAVDIASIHAQTLAATLPGEEAVREALRRAAEIIAEIEAE
part2_trp249_1p9m_9_model_0_1	0.77	17.03	4.04	-149.91	-20.7	-33.96	PEVRTETITSDLPPEEVRKEVAARIAELKAQGAKEVVVID QRVEGSSTVT TLEVTY
part2_trp249_1p9m_6_model_6_1	0.75	4.4	2.1	-119.54	-21	-34.72	RVVVEREVIEFEEPIIEEARPIAERRVEELERRLREEYPD AEIEVRASFIEEDGLVKELIVEAVVRVP
part2_trp249_1p9m_30_model_1_1	0.71	18.45	3.24	-133.3	-22.21	-39.46	ATAVVTVDSPRPRTVDIECVEDPGIRLRATYPPGRHERTF EVPPG SVIRFLDPANAEGVTVTVTE
part2_trp249_1p9m_39_model_0_1	0.72	16.42	1.32	-137.14	-24	-32.83	PRLRTLTYYPKEKAEERIRIVAAL EASGLRVVDVKREQ VDEDTV RITITVEE
part2_trp249_1p9m_33_model_9_1	0.77	14.91	7.33	-124.44	-24.7	-38.2	VTVRVGPFATEEEAVAAAEAGREVAARLGGR LAEIGLV RGEAWTGVAVVEVP

SynthImmunol_NMU			Notebook			No.5	
part2_trp249_1p9m_36_model_1_1	0.73	16.63	7.61	-175.33	-26.15	-34.45	RELEQVIVLPPDAELLSVSGDCTVEVEELPDGTTTRVTVRY REGQTFITITYRYRDDVVTETFSV
part2_trp249_1p9m_29_model_7_1	0.78	7.72	1.64	-183.09	-27.84	-44.56	DTLARQEAVEAFRLRARGLEPQQIELRLEPDGSWVAVAYV DAGPGRLLRVTVREVPGGGWEVVAEELLELP
part2_trp249_1p9m_31_model_0_1	0.73	23.23	4.13	-172.53	-28.95	-40.2	PCVLPPPFTLTVLFPGDAEPRTWLCTPVRCFTEDGVPLAE YACRSLEDPSLTGKAVLRLEGDEWVVSVSYNRPEP
part2_trp249_1p9m_31_model_8_1	0.74	20.85	5.04	-196.08	-29.22	-46.4	PGRTLITLVTVSGAESVEAARAAALAQLAALKPGTTIT IAGRPYRLLSVGEYTVIEDPKTKSATVEVEAVVQEL
part2_trp249_1p9m_16_model_5_1	0.72	23.32	8.89	-89.3	-29.49	-41.64	AAELRELAALVEAAFPEWRVVGFEEDGELCVVALREL PDGRLLRALFRRREDGSVELVRVEEVPL
part2_trp249_1p9m_7_model_3_1	0.72	19.1	3.57	-177.08	-30.3	-33.29	APRRYVITDDRELMEEILEADPSVVAITVESREAAERLAA RLRAVPEDVADVEVRVP
part2_trp249_1p9m_23_model_7_1	0.79	17.61	4.26	-125.66	-31.22	-42.88	MLKKYLVARPVVEGNQPSIEIEVEAASPEEAEKAIEILRE QGITGTWVISVIVDGELVERRTVTIE
part2_trp249_1p9m_12_model_4_1	0.78	5.27	1.52	-225.85	-36.16	-55.74	MTPEQLAKADEYFAKSEEARKKAGEYCLKLYDETGDPALLEKAE AYRKEADIYTAKGYKAAGDTETAKAIADKLGVTL
part2_trp249_1p9m_22_model_4_1	0.81	19.44	8.87	-162.69	-38.14	-43.07	GMKRMRETITVAPEKVPEAVALLEELKARLPEARIEFAVDGNTIT VEVEFPEEALPRLEEALAEFYARIRPLSRRREYES
part2_trp249_1p9m_38_model_8_1	0.78	5.66	1.32	-231.18	-39.91	-55.54	REEEFEALVAEHWPEIEAVYAEAARTLSREETAAAVRAAIEALLP PEVAAYAIAHYAPKLE
part2_trp249_1p9m_2_model_7_1	0.78	3.74	2.46	-237.48	-47.69	-61.14	MLKFLGLSITPAPQTEEEYEKVEKLQKEVEEKKHKELSKKLDKETT LKELKKYAEEKAKELKLPKRINIHVEEGE
part2_trp249_1p9m_10_model_5_1	0.78	9.15	1.77	-164.51	-49.62	-65.61	MEVVERTLEIDAAAPEEEAVARVLAALPALPEALARAAVREVKA NPGSEVTVGvetPERRELLTVVSR
part2_trp249_1p9m_19_model_9_1	0.82	8.27	1	-238.06	-56.44	-62.69	RLDELVASAAHLAELGLAPEEVAAAVARIRAAAAAARAANAS LPDPRLRERAVLDAASAEIERIGRERAA
part2_trp249_1p9m_6_model_2_1	0.74	15.29	1.78	-192.2	-60.73	-72.8	SSKVLRYRRVGDADIDELIATVKEQLAAEGASDIRLLERSPDGKR VVLVYKDPNGEERYAIVEVVVEEP
part2_trp249_1p9m_35_model_5_1	0.76	18.32	1.63	-326.46	-62.66	-74.16	SAALRARGLAALRAEAERLGLDPAAVRLVAVLRVGDRVVALGE LPPEAYPHADPDDSRVRFEAEVPA

SynthImmunol_NMU			Notebook			No.5	
part2_trp249_1p9m_31_model_3_1	0.78	20.29	2.86	-127.65	-64.53	-59.88	IPANVQVGLPDPSLLPLALKAVEVARKVVGPDATIGVTYNPETKT FSVHAQAPGQPPLYLIVAEDGTVLEEGERVPP
part2_trp249_1p9m_5_model_3_1	0.64	1.67	1.76	-274.29	-67.9	-69.43	SVKVVTSQCTLPGDTVVTVTVSCTGPGEYTWESFYDADDDPSNR TPPCTVTCTCSPDDP
part2_trp249_1p9m_6_model_7_1	0.8	10.14	1.7	-231.37	-70.53	-83.02	APSAEQLAALAAEARAALLAQAAELGWTPSDLELEVRPVPEGVL AIARATATRPGQPPVRLRAEALLRL
part2_trp296_1p9m_38_model_5_1	0.72	9.53	3.32	-128.66	-24.38	-30.51	ARHTLPLTVPEGETDVYVDGKLVATVPAGEQEVTVEVPPGTHRV SLRNRKTGESRSSDVVFP
part2_trp296_1p9m_25_model_8_1	0.71	22.82	8.66	-164.27	-32.59	-44.19	GAAARQAAALDAAAEAAEAAGDLAAAAAYAAAAALLAAHA ATHPNGANLAALAEAAARAAALAALAA
part2_trp296_1p9m_29_model_0_1	0.72	22	3.84	-198.33	-65.43	-74.04	GFRAAVPLRVGDPVEIRLDDGTVVPGRVDRVEGDVVSVAPEPL TPGVHRVTVRFLTEPPQTSEPRLVEFPVRVP

After sorting by the average score per residue and selecting the top 1 sequence, a total of 78 candidate sequences were obtained. ID, binder identifier. Binder PTM, predicted post-translational modification score. Minimum inter-chain PAE, minimum inter-chain predicted aligned error. Target-aligned binder RMSD, RMSD of the binder aligned to the target (Å). Statistical potential, statistical potential score of the complex. MM/GBSA and MM/PBSA, binding free energies calculated by MM/GBSA and MM/PBSA methods (kcal/mol), respectively. Binder Seq, amino acid sequence of the designed binder.

6. AI Use Statement:

Computational prediction generated using AI, not experimentally validated by wet experiment. We employed a combined approach of computational design and experimental verification. Therefore, for results from AI-assisted prediction/simulation, we will continue to conduct wet experimental verification to confirm their validity.